## 4.4 Rotational Symmetry

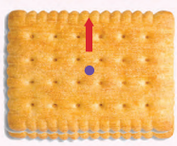
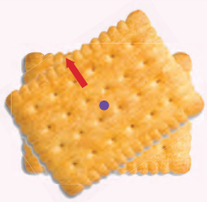
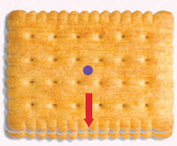
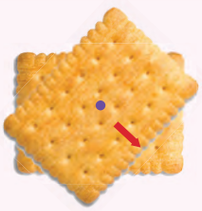
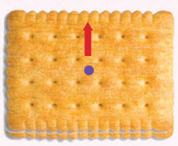
We have already learnt about rotation. **Rotation** means turning around a centre. The paper windmill, merry-go-round, fan, tops, wheels of vehicles, fidget spinner are few examples of rotating objects that we see in our life.

| | | | |
|---|---|---|---|
|  |  |  |  |
| Paper wind mill | Wheels | Fan | Merry-go-round |

When one rotation is completed, the rotating object comes back to the position where it started. During a complete rotation, the object moves through 360°.

### Think about the situation

1) Take two rectangular biscuits from the same packet and put one on the other. Holding one biscuit firmly rotate the other on it about the centre.

| | | | | |
|---|---|---|---|---|
|  |  |  |  |  |
| Initial position | Rotation | First Match | Rotation | Second Match |

How many times does it fit exactly on the other in a complete rotation? Two times.

2) In the example given below, if you rotate the fidget spinner about the centre, there are three positions in which the fidget spinner matches exactly the same in a full rotation.

| | | | | | | |
|---|---|---|---|---|---|---|
|  |  |  |  |  |  |  |
| Initial position | Rotation | I-Match | Rotation | II-Match | Rotation | III-Match |

3) Place a set square (containing angles 60°, 30° and 90°) on a paper and draw an outer line around it. What type of triangle do you get? Yes, Scalene triangle. If you rotate it about the centre, there is only one position in which the set square fits exactly inside the outer line.



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